Band-1 Computational Experiment: Goldbach Verification — Explicit Range Collapse

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Goldbach Verification (Range Collapse)

Honesty stamp. This note **verifies** that every even integer n with $4 \le n \le X$ is a sum of two primes for an explicit finite X computed on this box. It does **not** prove the **strong Goldbach conjecture** (every even integer ≥ 4 is a sum of two primes). Finite-range verification $\neq$ conjecture. The standing **peer-reviewed** verification record remains $4 \cdot 10^{18}$ (Oliveira e Silva–Herzog–Pardi, *Mathematics of Computation* 83 (2014), 2033–2060). Preprint / Zenodo claims beyond $4 \cdot 10^{18}$ are ledger-noted but **not** elevated above the published Math. Comp. record.

Abstract

We implement a **segmented sieve of Eratosthenes** plus a minimal-partition scan and verify the even Goldbach property on $[4, X]$ with

$$\mathbf{X = 200,000,000 = 2 \cdot 10^8}.$$

All **99,999,999** even integers in the range admit a Goldbach partition (failures: **0**). Wall time **107.35 s** (sieve 14.19 s + verify 92.73 s). This is a Band-1 **range-collapse / methods** certificate at a historically meaningful scale (beyond Shen's 1964 10^8 milestone; far below the published $4 \cdot 10^{18}$ record). **Conjecture status: OPEN.**

Problem statement

Strong (even) Goldbach conjecture. Every even integer $n \ge 4$ is the sum of two primes.

This experiment. Prove computationally that the statement holds for all even $n \in [4, X]$ for a chosen explicit X tractable on this box (minutes), and record the literature ledger of larger verified ranges.

Literature ledger (not re-proved on box)

Year	Limit	Authors	Status
1855	10^5	Desboves (Haussner 1896)	historical
1940	$3.3 \cdot 10^7$	Pipping	historical
1964	10^8	Shen	historical
1965	$2 \cdot 10^{10}$	Stein & Stein (Light et al. 1980)	historical
1989	$4 \cdot 10^{11}$	Granville, van de Lune, te Riele	published

Year	Limit	Authors	Status
1993	10^{14}	Sinisalo	published
1998	10^{14}	Deshouillers, te Riele, Saouter	published
2001	$4 \cdot 10^{14}$	Richstein, *Math. Comp.* 70	published
2014	$4 \cdot 10^{18}$	**Oliveira e Silva, Herzog, Pardi**, *Math. Comp.* 83	**published record**
2025	beyond $4 \cdot 10^{18}$ (partial)	various preprints / Zenodo	**not** peer-reviewed record

Primary citation for the published record: T. Oliveira e Silva, S. Herzog, S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to* $4 \cdot 10^{18}$, Mathematics of Computation **83** (2014), 2033–2060 (DOI 10.1090/S0025-5718-2013-02787-1). Author page: https://sweet.ua.pt/tos/goldbach.html.

Odd Goldbach (every odd > 5 is sum of three primes) is a theorem for large n (Helfgott); OHS note that even-verification to $4 \cdot 10^{18}$ + Ramaré–Saouter implies odd Goldbach up to $8.37 \cdot 10^{26}$ — recorded for context only.

Method (box)

1. **Segmented sieve.** Generate all primes $p \le X$ via a segmented sieve of Eratosthenes (base primes $\le \sqrt{X}$; mark composites in segments).
2. **Primality bitset.** Boolean array `is_prime[0..X]` for $O(1)$ lookups.
3. **Minimal partition scan.** For each even $n = 4, 6, \ldots, X$, find the smallest prime $p \le n/2$ such that $n - p$ is prime. Record failures (expect none) and the maximum such minimal p over the range.
4. **Unit tests.** $\pi(10^6) = 78498$; sieve vs segmented agreement to 10^4; full verify on $[4, 30]$.

Complexity. Sieve $\tilde{O}(X \log\log X)$; verification typically near-linear in X because minimal Goldbach primes stay small (early exit).

Results (numbers!)

Unit tests

`PASS`: primes to 30; verify to 30; 28 has a partition; sieve≡segmented at 10^4; $\pi(10^6) = 78498$.

Box verification

Quantity	Value
X **verified**	**200,000,000** $(2 \cdot 10^8)$
Even integers checked	99,999,999
Failures	**0**
All verified	**True**
$\pi(X)$	11,078,937
Max minimal-prime p	**1321** at $n = 187,852,862$
Sieve time	14.1880 s
Verify time	92.7272 s
Wall total	**107.3471 s**

Sample minimal partitions

n	p	$n-p$
4	2	2
10	3	7
100	3	97
1000	3	997
10000	59	9941
100000	11	99989
1000000	17	999983
10000000	29	9999971
100000000	11	99999989
200000000	37	199999963

Calibration

Auto-selected X from ladder $\{10^6, 10^7, 5\cdot 10^7, 10^8, 2\cdot 10^8, 5\cdot 10^8\}$ under a ~3-minute wall budget (estimate model $T \propto X \log X$ from a 10^6 probe).

Why Band-1 vs Band-4

- **Band-1 (this note):** shrink / certify an **explicit finite range**; exercise the classical segmented-sieve + partition methodology used in the literature at world-record scale; ledger hygiene on published vs preprint claims.
- **Band-4:** a proof that **every** even integer ≥ 4 is a sum of two primes. Still **OPEN**; **not claimed**.

Reproducibility

- Code: `/workspace/experiments/band1/goldbach_verify/`
- Run: `python3 goldbach_verify.py`
- Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_goldbach_verify.log`

Conclusion

Box: every even integer in $[4,\ 2\cdot 10^8]$ is a sum of two primes (99,999,999 checks; 0 failures; ~107 s). **Literature:** published verification record remains $4\cdot 10^{18}$ (OHS 2014). **Live Band-1 target:** larger explicit X and ledger hygiene — **without claiming the Goldbach conjecture is solved.**